

Molecular Physiology of Vascular Smooth Muscle Cells

Principles and Practices of Osteopathic Medicine II



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Do.
Make.
Heal.

Innovate.
Reinvent the Future.

Learning Objectives

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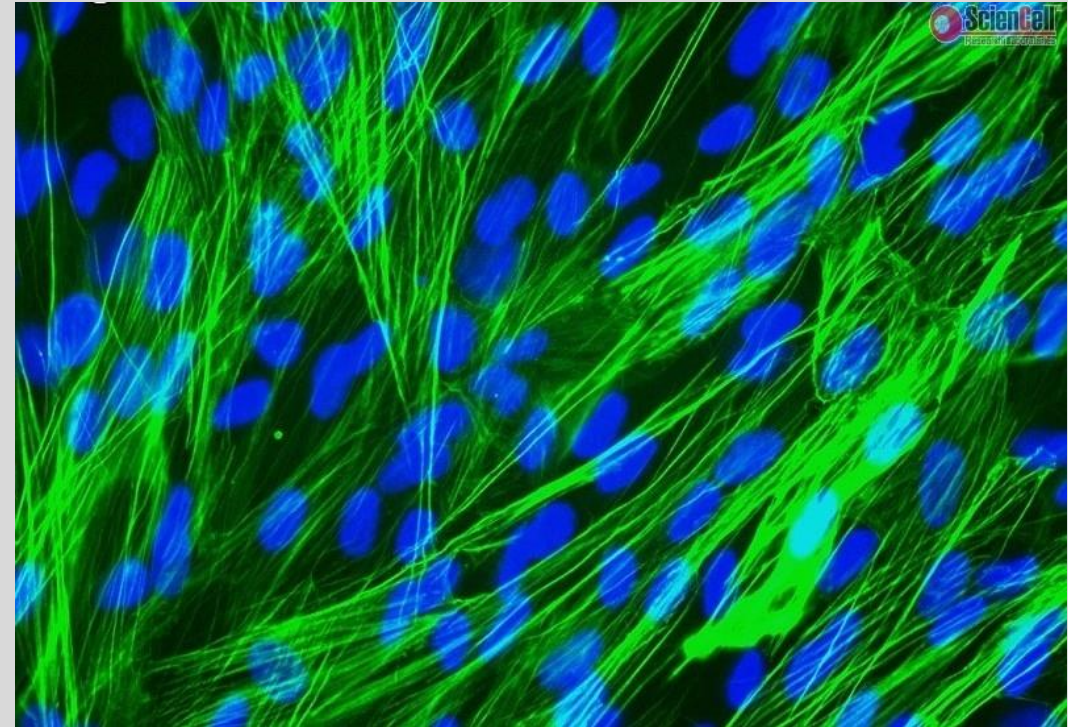
- Learning Objective 1: Describe the main mechanisms by which VSMC contracts and relaxes
- Learning Objective 2: Identify contractile proteins from VSMC and describe the source and movement of Ca^{2+} ions in the vascular smooth muscle cells.
- Learning Objective 3: Explain the mechanisms by which KCl induces vascular contraction
- Learning Objective 4: Recognize major phenotypic changes in diseased VSMC

Vascular Smooth Muscle Cells (VSMC)

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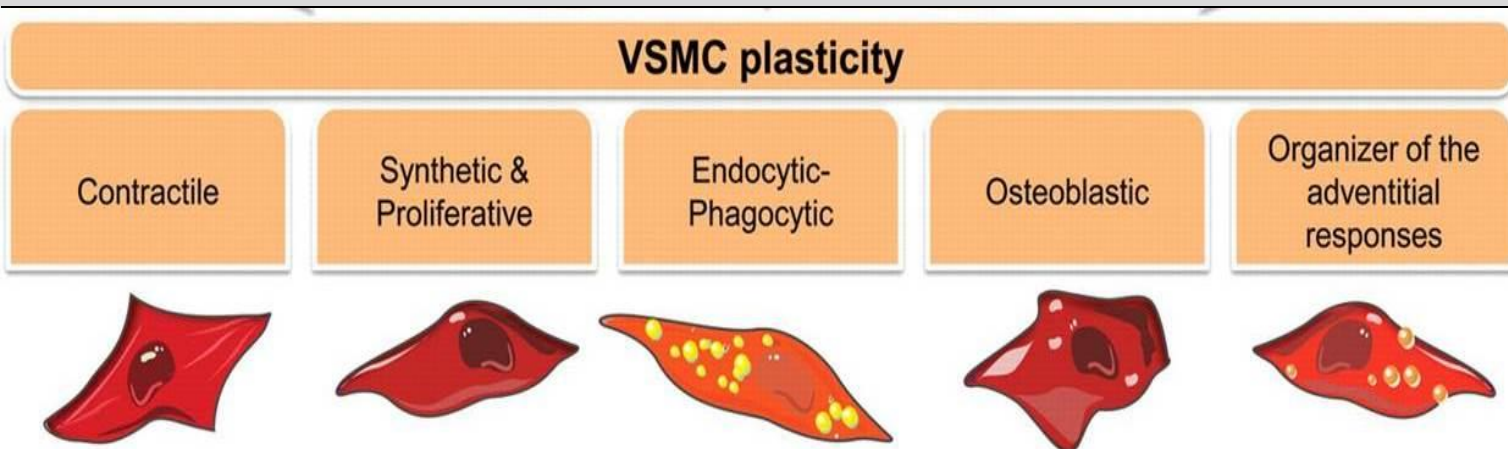
Spindle-shaped cells with singles nucleus



Images from Sepulveda's Laboratory- NYITCOM

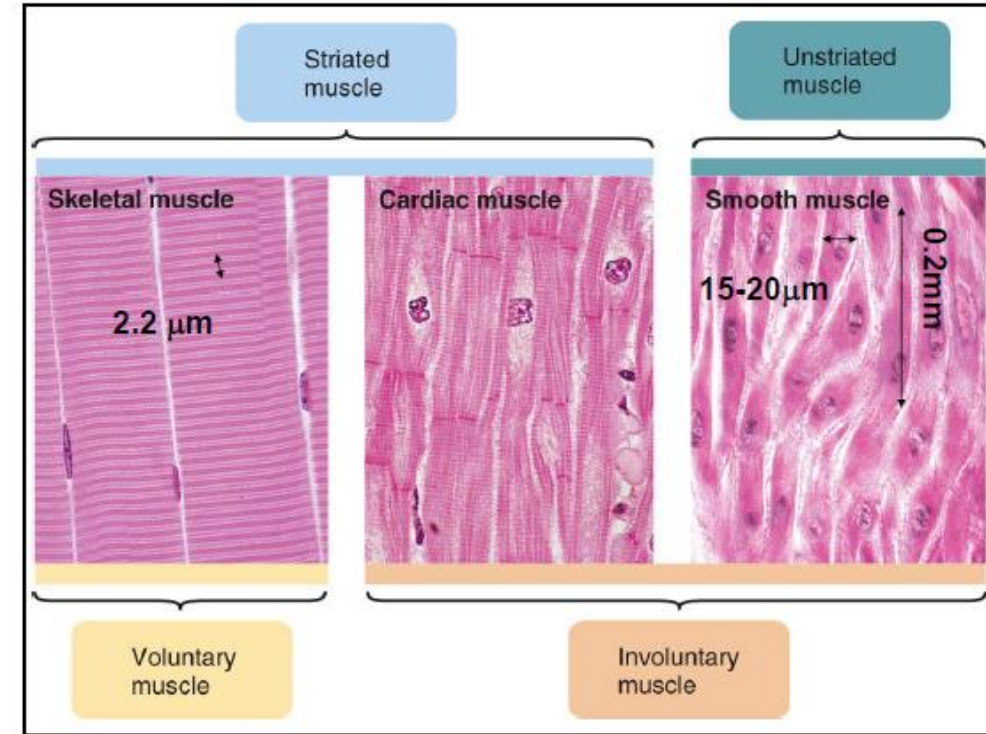
Basic Characteristics of VSMC

- ♥ Not striated
- ♥ Not organized in sarcomeres
- ♥ Troponin is absent. VSMC has calmodulin
- ♥ **Native contractile phenotype**
- ♥ **Dense bodies** instead of Z disks



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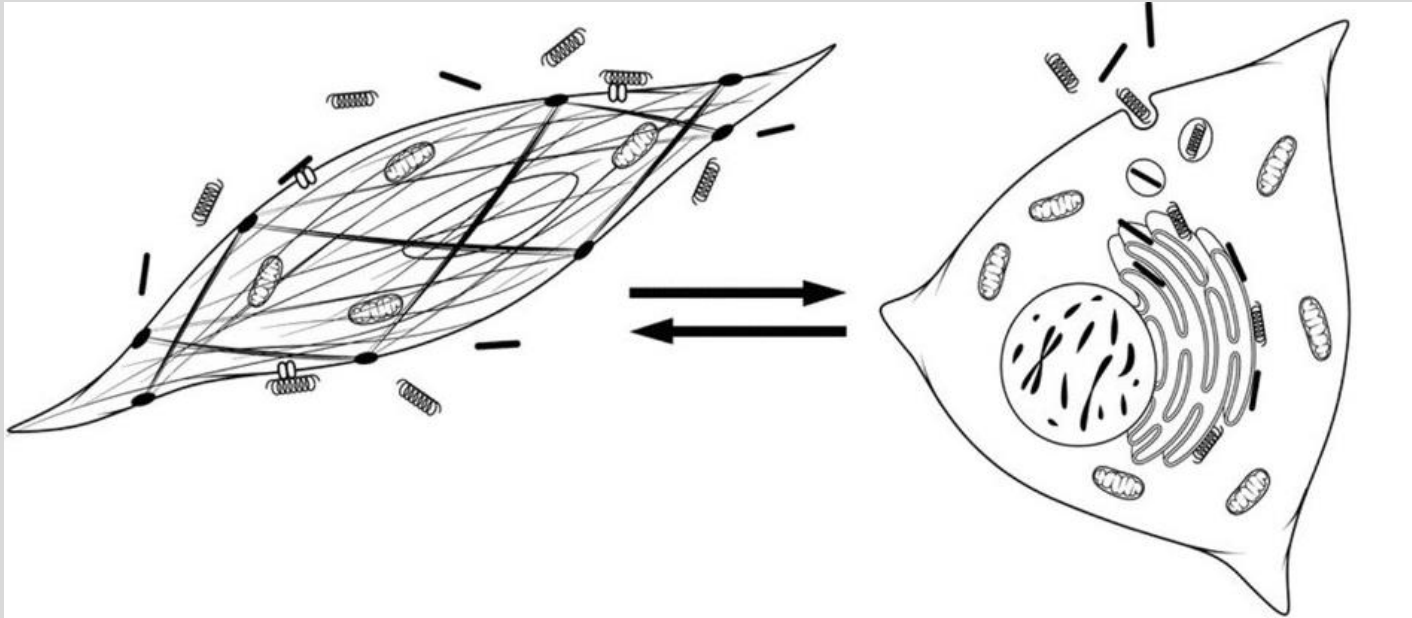
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Normal and Diseased VSMC

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Normal

Spindle shape

Contractile

Nonproliferating

Nonmigrating

Diseased

Rhomboid shape

Secretory/Synthetic (synthesizing
extracellular matrix)

Proliferating (cell division)

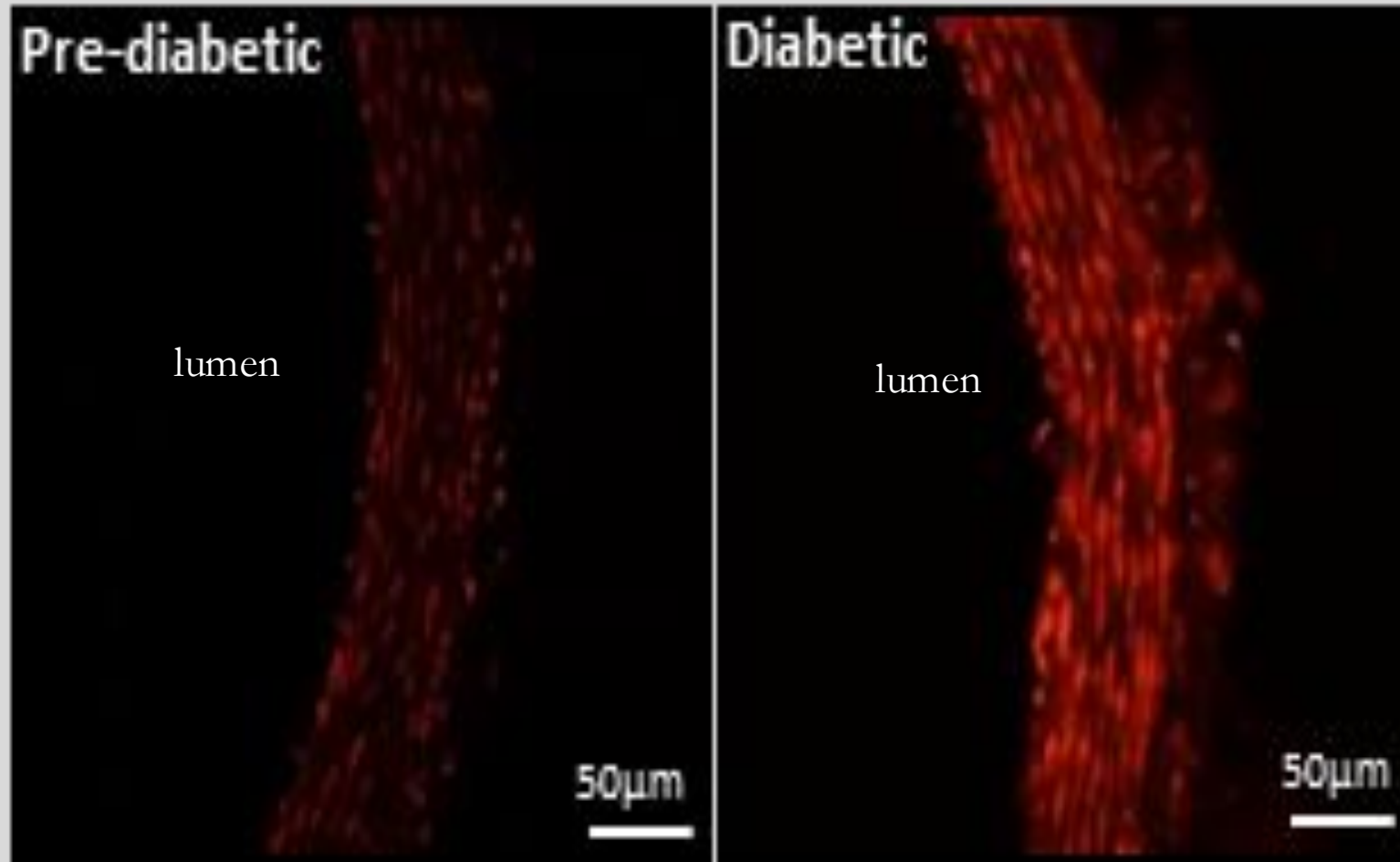
Migrating (able to invade)

Normal and Diseased VSMC

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Red: staining for Reactive
oxygen species (ROS)

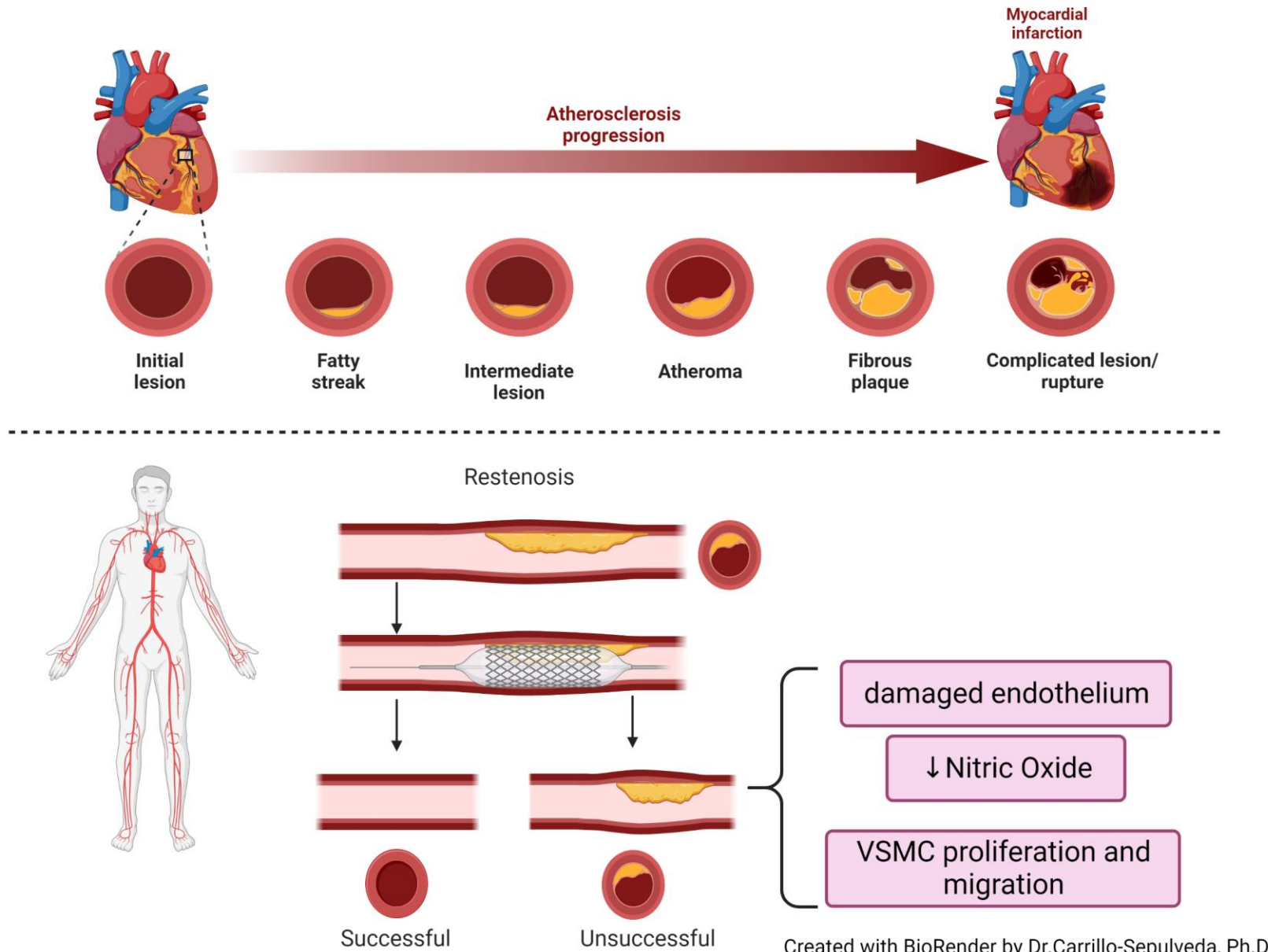


Images from Sepulveda's Laboratory- NYITCOM

Normal and Diseased VSMC

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VSMC in Atherosclerosis

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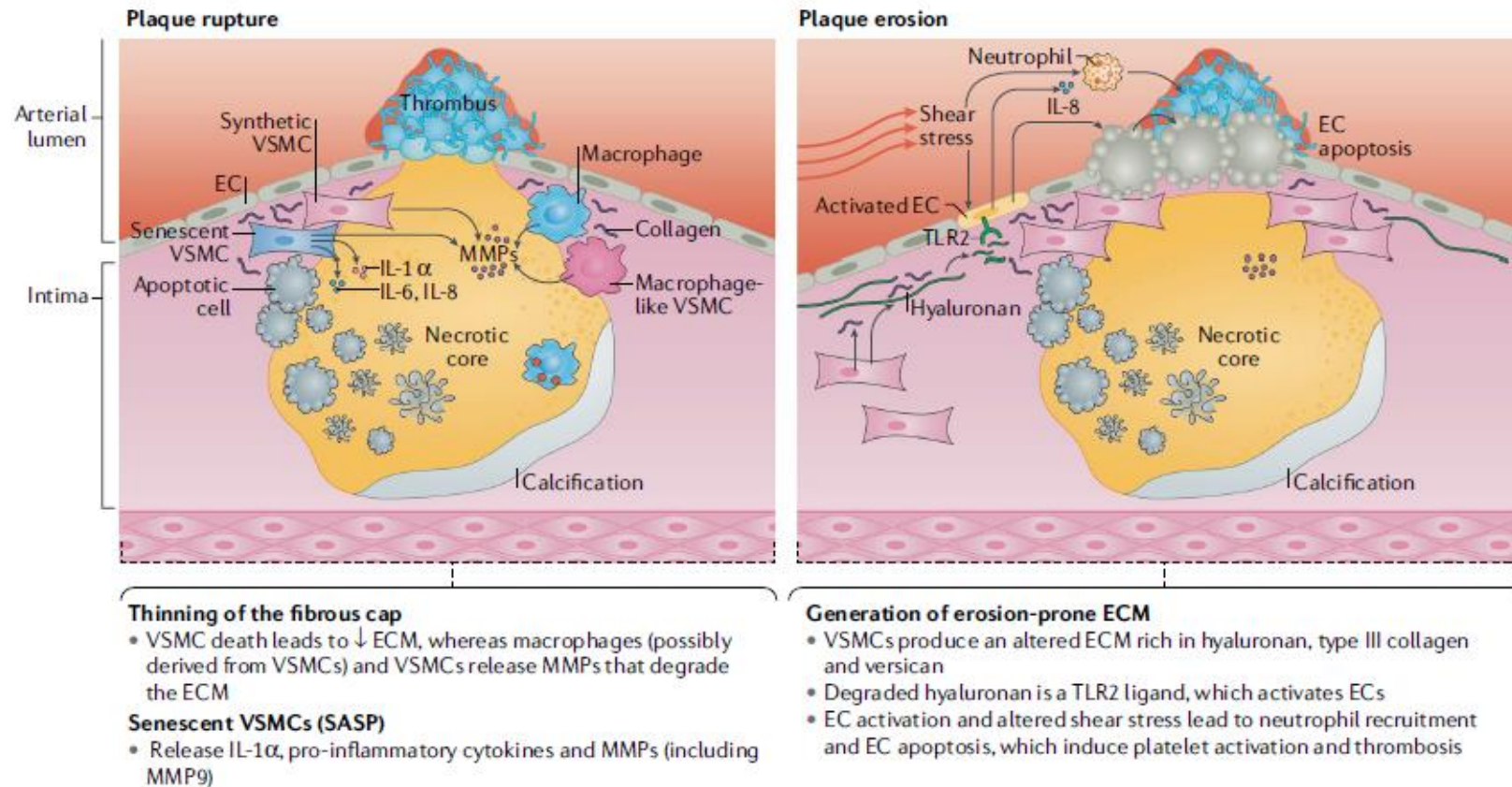


Fig. 4 | **VSMCs in clinical sequelae of atherosclerosis.** Summary of the role of vascular smooth muscle cells (VSMCs) in atherosclerotic plaque rupture and plaque erosion, the two major processes underlying thrombosis and, thereby, the clinical sequelae of atherosclerosis. EC, endothelial cell; ECM, extracellular matrix; MMP, matrix metalloproteinase; SASP, senescence-associated secretory phenotype; TLR, Toll-like receptor.

Practice Question

A 59-year-old man becomes unconscious after experiencing severe retrosternal chest pain. He has a history of smoking and dyslipidemia but is noncompliant with his prescribed medication regimen. The man is pronounced dead on arrival to the emergency room. An autopsy reveals diffuse atherosclerotic vascular disease involving the coronary arteries. One of the atheromatous plaques in the left main artery has a ruptured fibrous cap with an overlying occluding thrombus. Which of the following cellular components is most directly involved in formation of the fibrous portion of this patient's plaque?

- a) Endothelial Cells
- b) Activated Platelets
- c) Smooth Muscle Cells
- d) Macrophages
- e) Infiltrating T lymphocytes

Structure of VSMC

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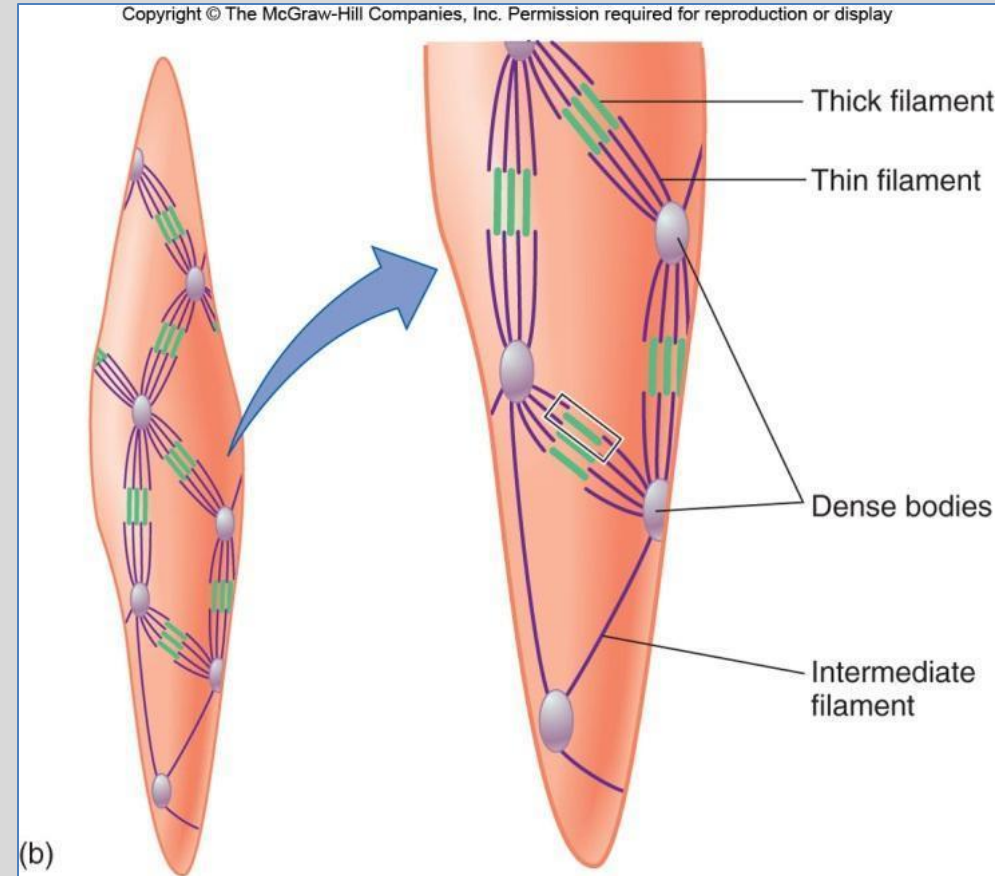
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Cytoskeletal filaments

1. Thin filaments : Actin (**purple**)
2. Thick filaments: Myosin (**green**)
3. Intermediate filaments: Desmin and Vimentin. Support the shape of the smooth muscle cells and hold the dense bodies in place

Dense bodies

Located in the cytoplasm and at the cell membrane.

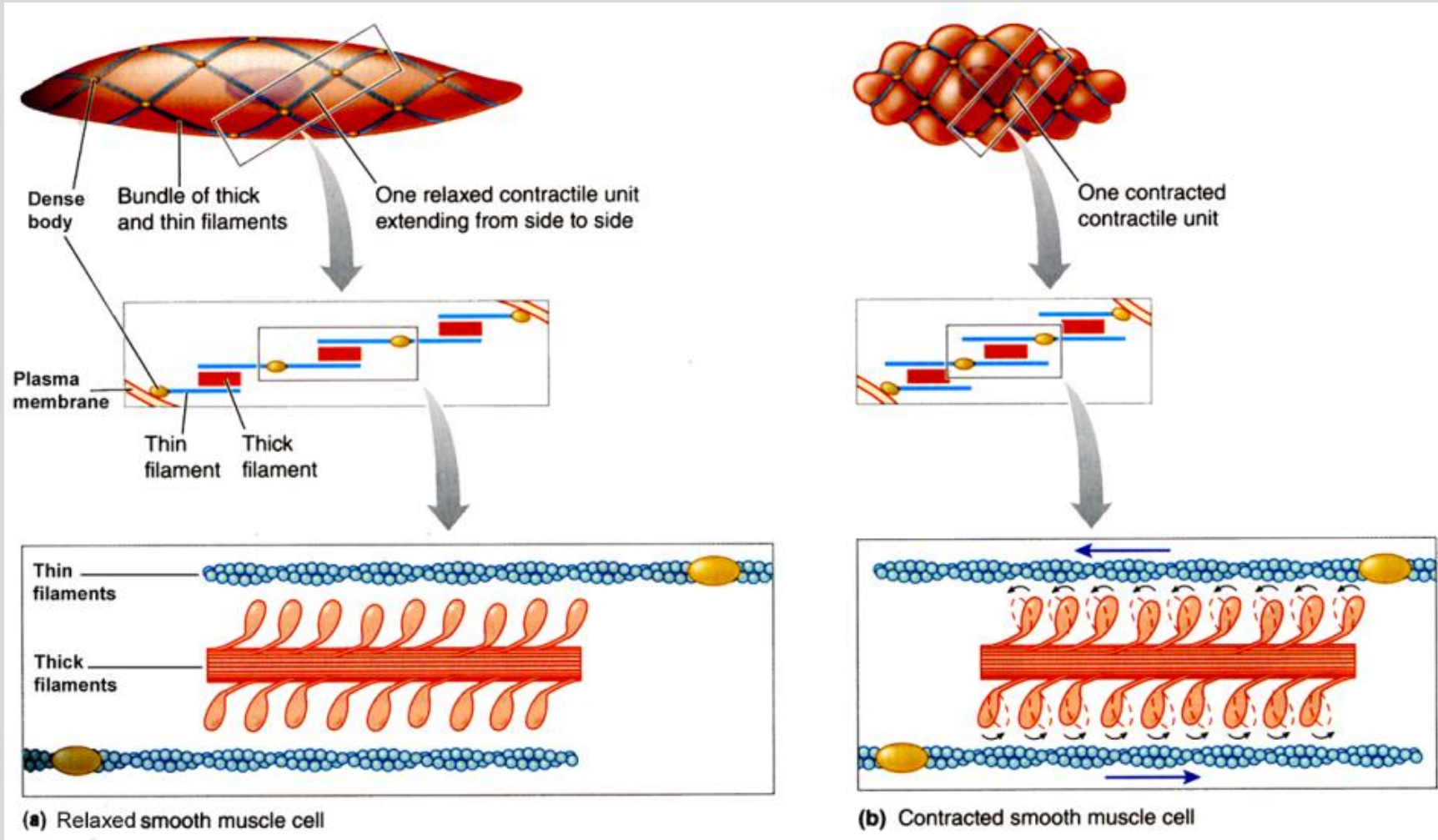


Sherwood, Human Physiology: From Cells to Systems, 8th ed., Brooks/Cole, 2013
(Chapter 8: pp. 294-295)

Arrangement of Contractile Proteins

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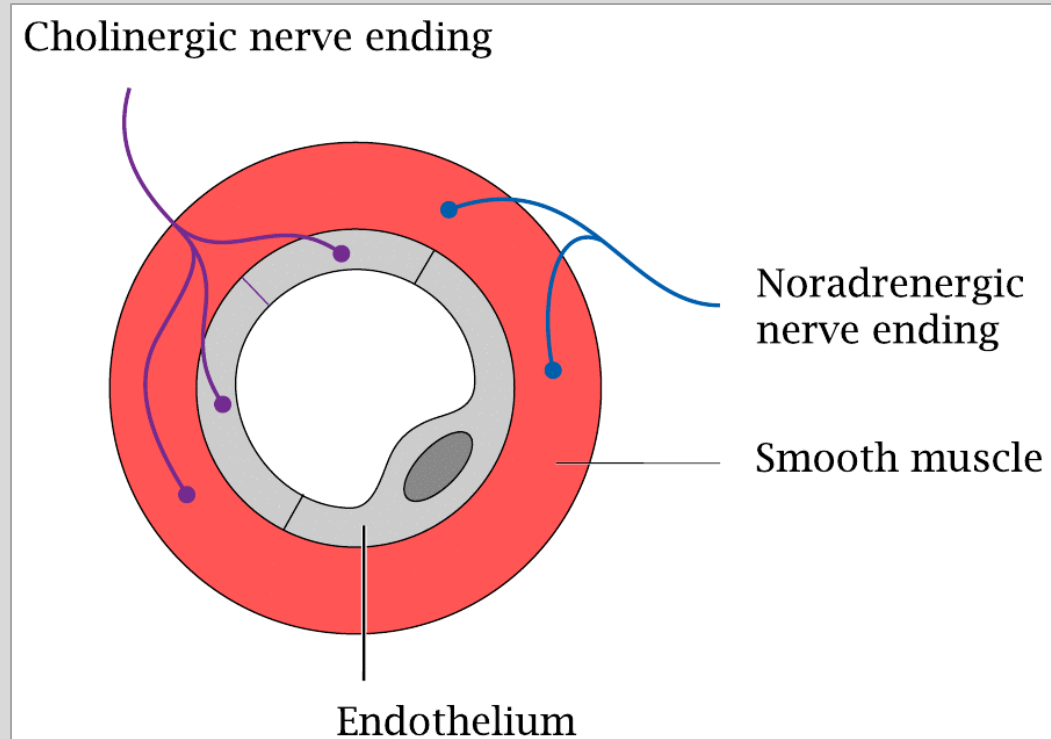
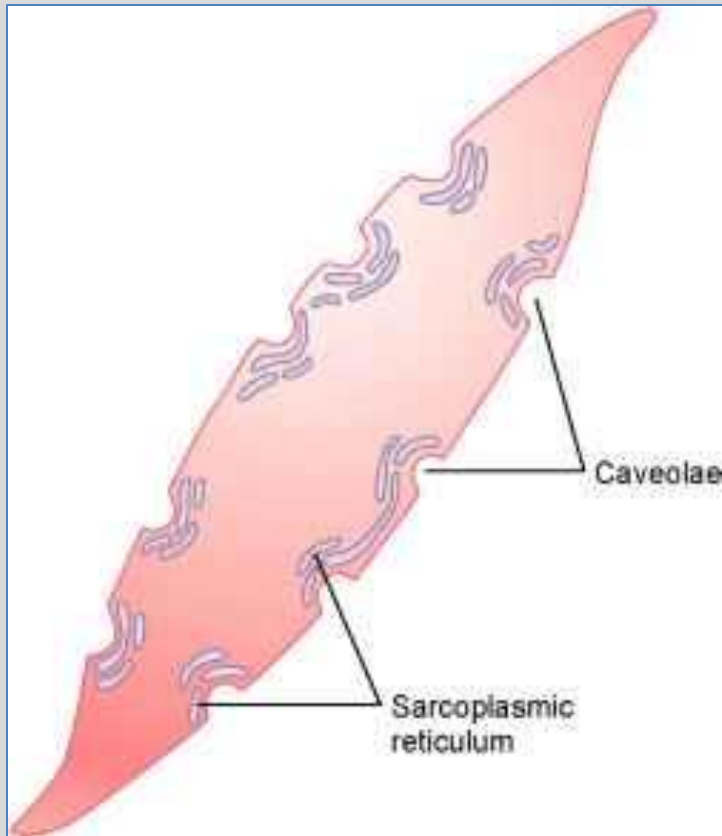


Structure of VSMC

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Caveolae

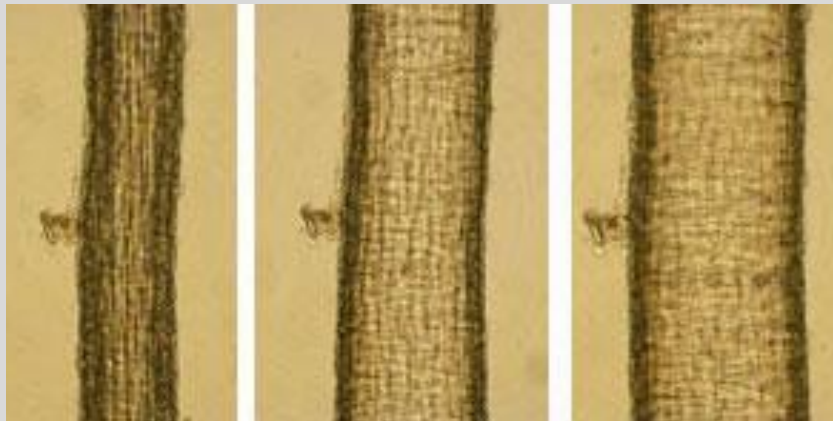


Smooth Muscle

- It is an involuntary muscle because it is controlled by **autonomic nervous system**.
- Smooth muscle can remain contracted for long periods (latch state);
- No fatigue
- It is able to initiate its own contraction, but neurotransmitters and hormones can change the rate and strength of muscle contraction.

Myogenic Tone

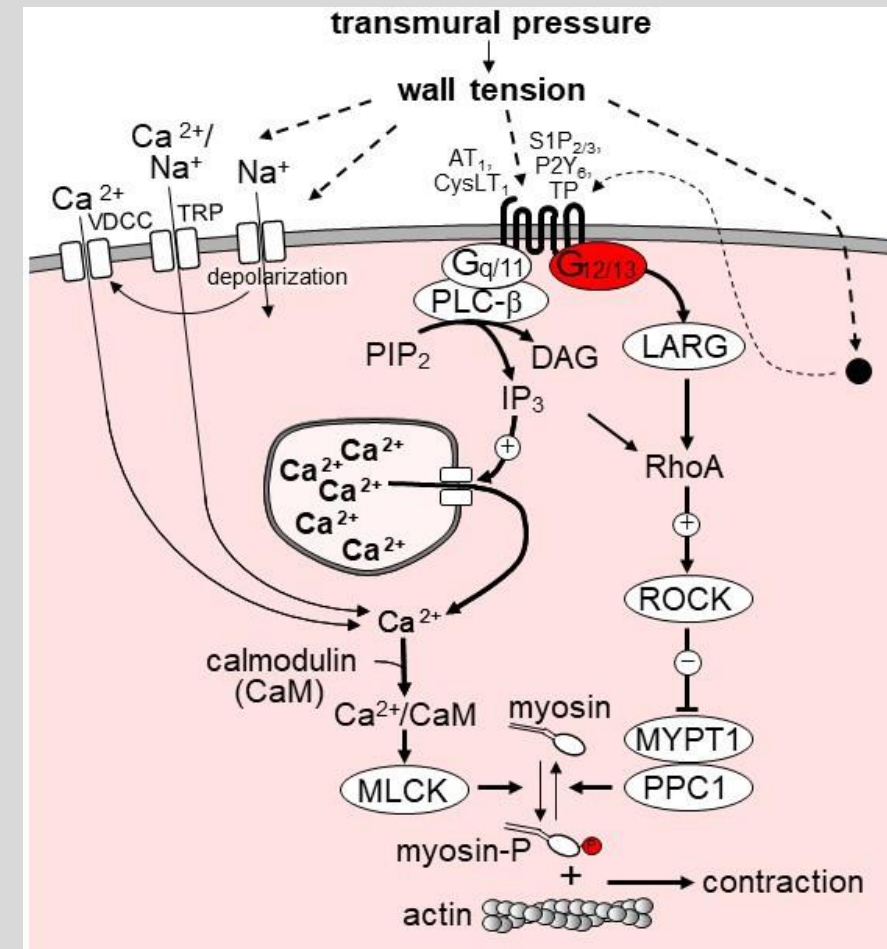
- It is a spontaneous vascular tone that originates from the muscle itself rather than from the autonomic nervous system or from hormones processes
- Small caliber blood vessels constrict and relax in response to changes in intraluminal pressure.
- $\uparrow$ intraluminal pressure cause vasoconstriction
- $\downarrow$ intraluminal pressure cause vasodilation



Mesenteric arteries
second order

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Neural Control of VSMC Contraction

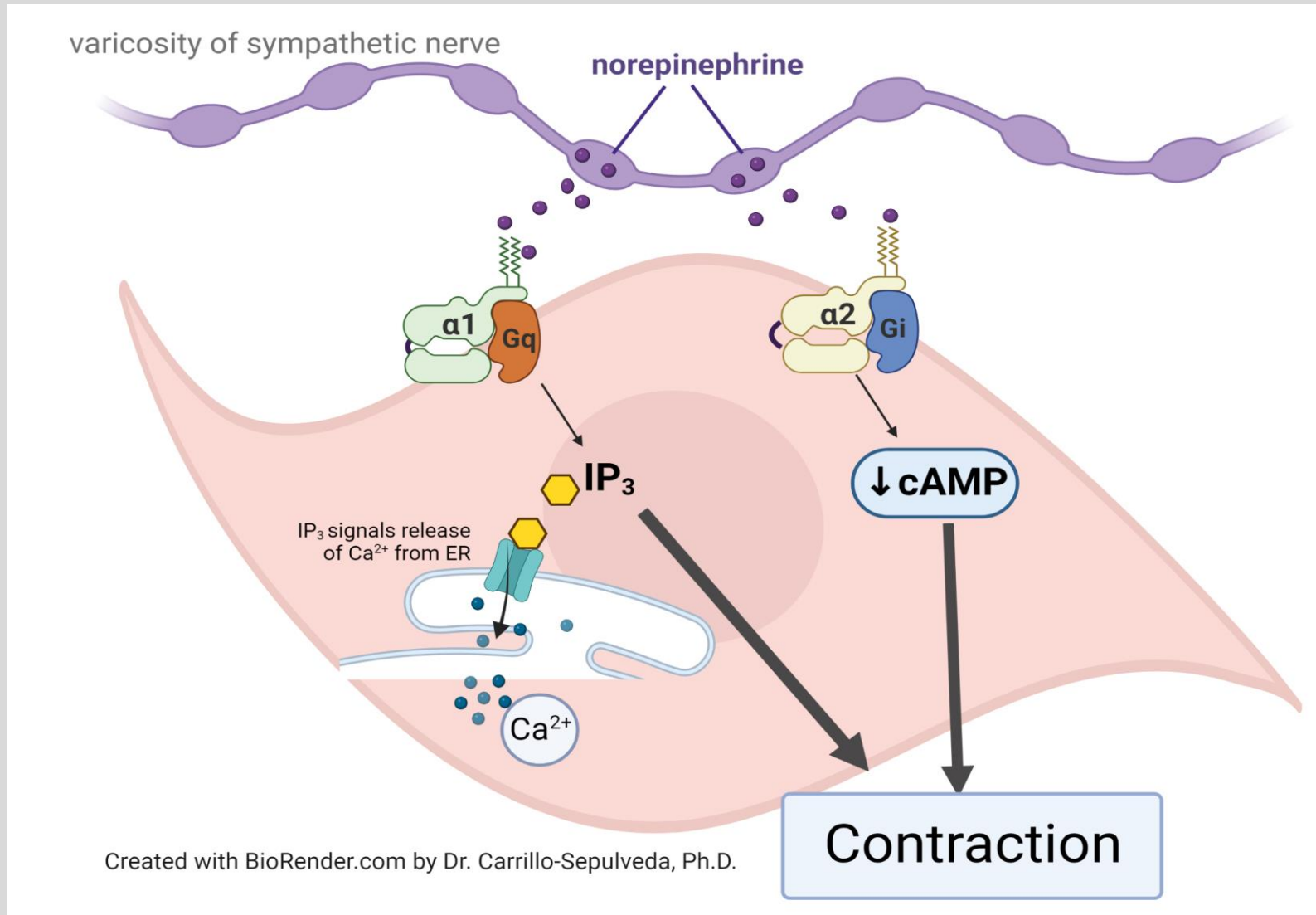
Neurotransmitter released by:

- Sympathetic Fiber: Norepinephrine. Alpha 1 and Alpha 2 adrenergic receptors

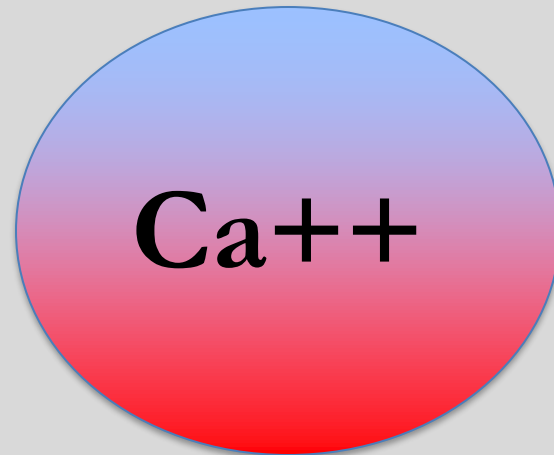
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SR instead of ER



After increases of calcium cytosolic, what is the next step of vascular smooth muscle contraction?

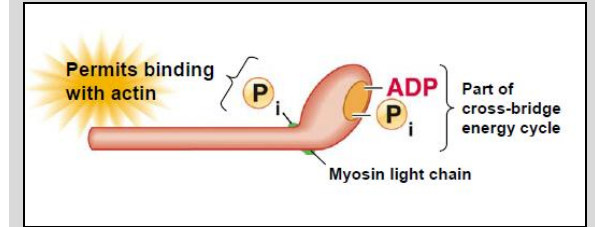
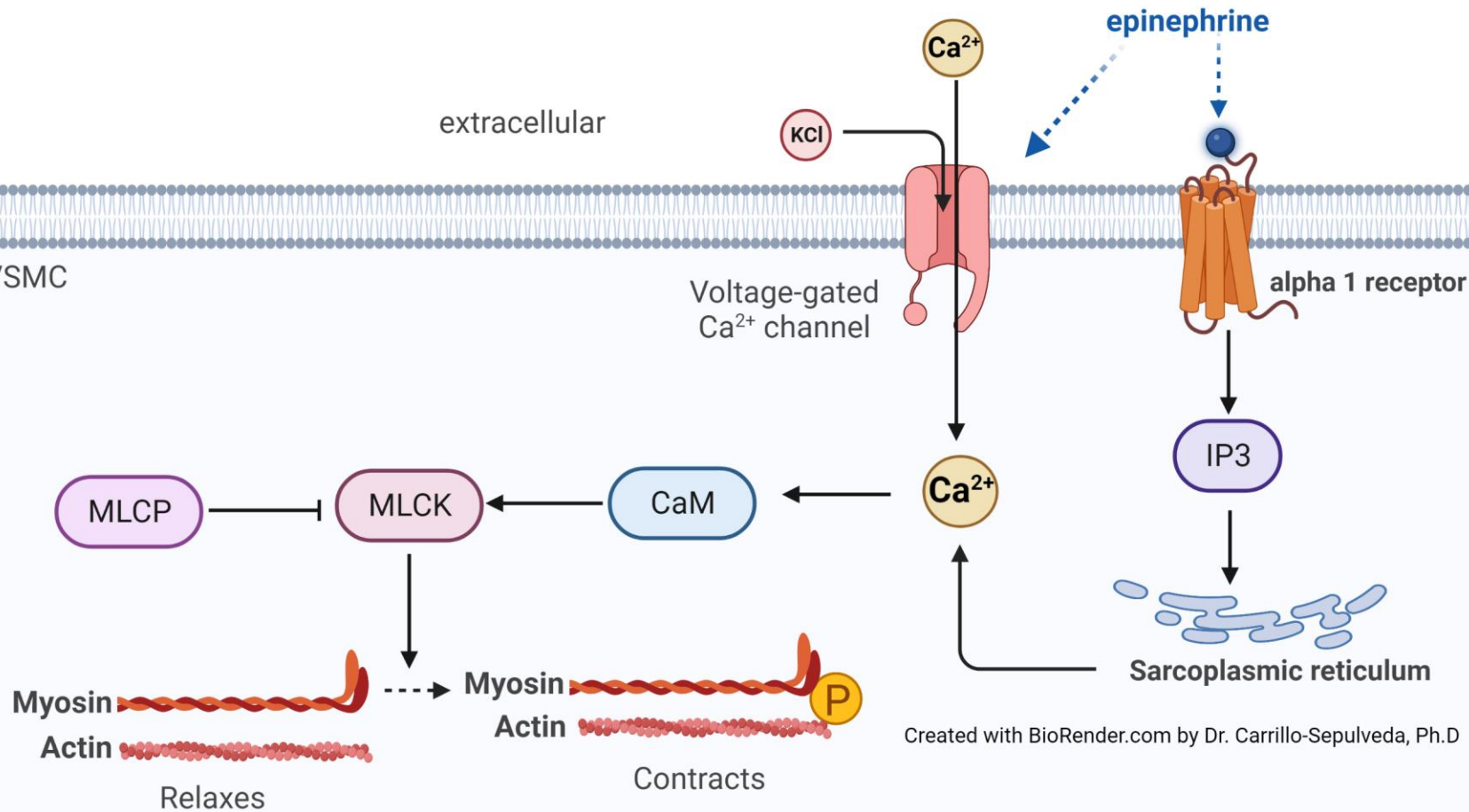


Vascular Smooth Muscle Contraction

Chemical event

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Unlike myosin light chain kinase, which is only active when cytoplasmic Ca^{2+} is elevated, *myosin light chain phosphatase is **always** active.*

Mechanism of VSMC Contraction

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↑Cytosolic Ca^{++}



Ca^{++} + Calmodulin



Activation of myosin kinase



Phosphorylation of myosin light chain



Attachment of myosin head with actin



Contraction by sliding filament mechanism



Contraction continues as
along as myosin is
phosphorylated

Practice Question

You are working in a scientific project which aim to develop a new vasodilator that can replace nitroglycerine. This new drug will be used to treat elevated blood pressure (hypertensive emergency). Which of the following intracellular events is most likely responsible for the beneficial effects of this “new vasodilator”?

- a. Actin phosphorylation
- b. Calcium release from sarcoplasmic reticulum
- c. Tyrosine kinase activation
- d. Myosin dephosphorylation

Ca⁺⁺ Source in VSMC

- Smooth muscle uses Ca⁺⁺ from both extracellular fluid and the sarcoplasmic reticulum

Source of extracellular Ca²⁺ ions :

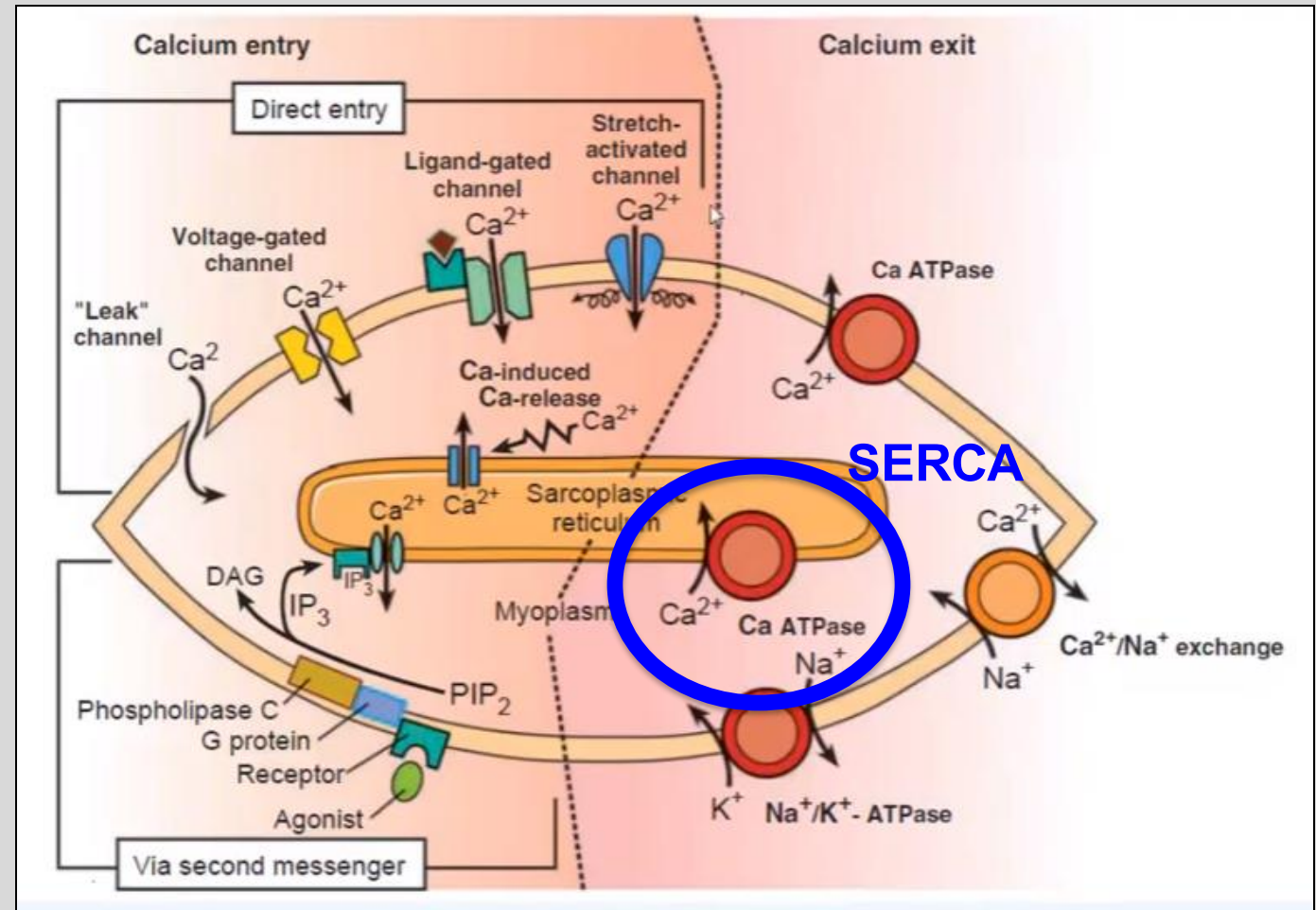
- Voltage-gated L-type Ca²⁺ channels
- Voltage-independent Ca²⁺ channels (ligand-gated, stretch activated channel, and “leak” channels)

Source of intracellular Ca²⁺ ions:

- Ca²⁺-induced Ca²⁺-release mechanism
- Inositol 1,4,5-triphosphate (IP₃)-mediated Ca²⁺ release

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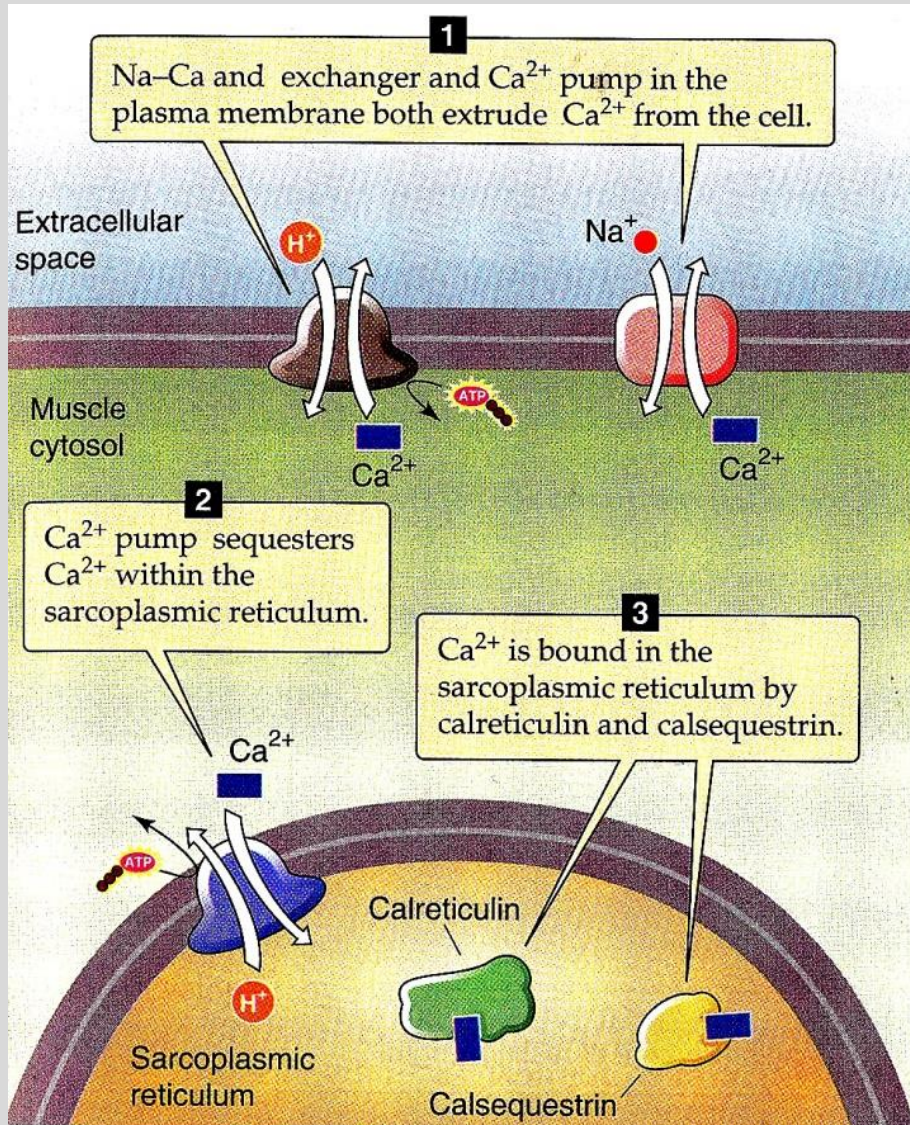
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Ca²⁺ Removal from VSMC

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Smooth muscle cell remove cytosolic Ca²⁺ by:

- ♥ Na⁺/Ca²⁺ exchanger in the plasmatic membrane
- ♥ Ca²⁺ pump in the plasmatic membrane.
- ♥ Ca²⁺-ATPase in sarcoplasmic reticulum (SERCA)

- Ca²⁺-binding proteins, such as calsequestrin and calreticulin, help to store Ca²⁺ ions in the sarcoplasmic reticulum.

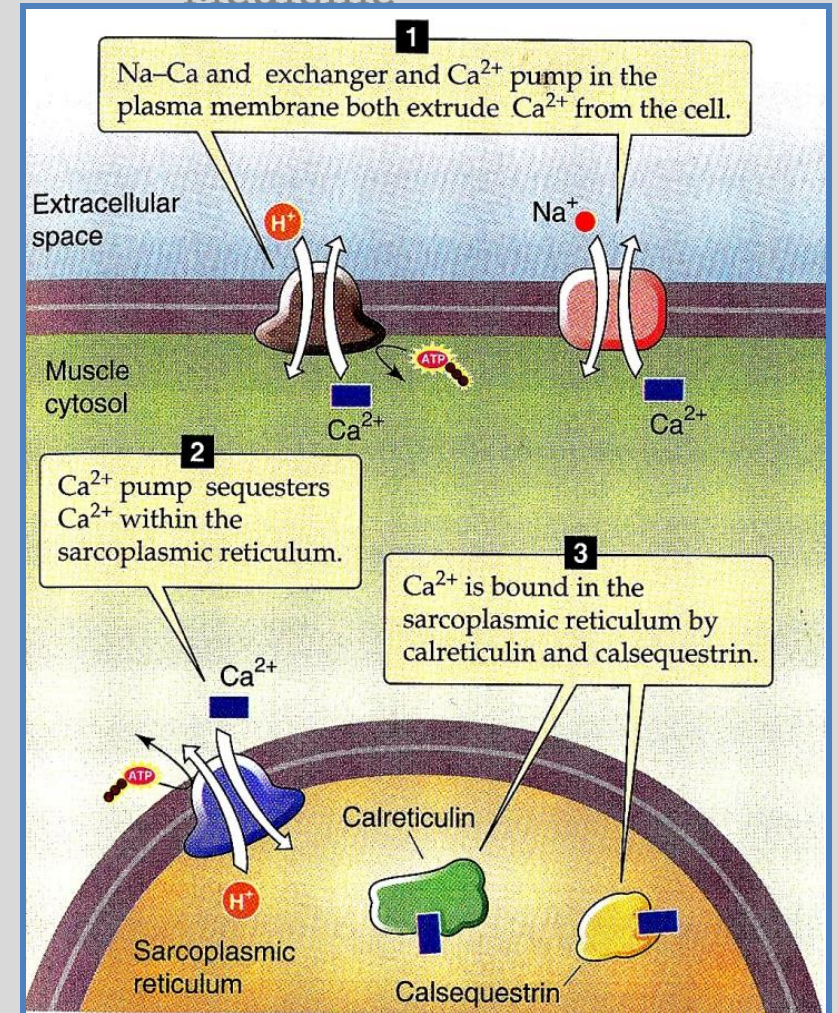
Vascular Smooth Muscle Relaxation

After a contraction-activating stimulus (electrical or chemical) has subsided, *Ca²⁺ ions must be removed from the cytoplasm for muscle relaxation to occur.*

- Extracellular extrusion of Ca²⁺ by Ca²⁺ pump and Na⁺/Ca²⁺ exchanger
- Ca²⁺ re-uptake in sarcoplasmic reticulum by Ca²⁺-ATPase (SERCA)
- Intracellular Ca²⁺ concentration falls to resting level
- Formation of Ca²⁺-calmodulin complexes decreases
- MLCK activity decreases
- MLC phosphatase dephosphorylates MLC on myosin heads

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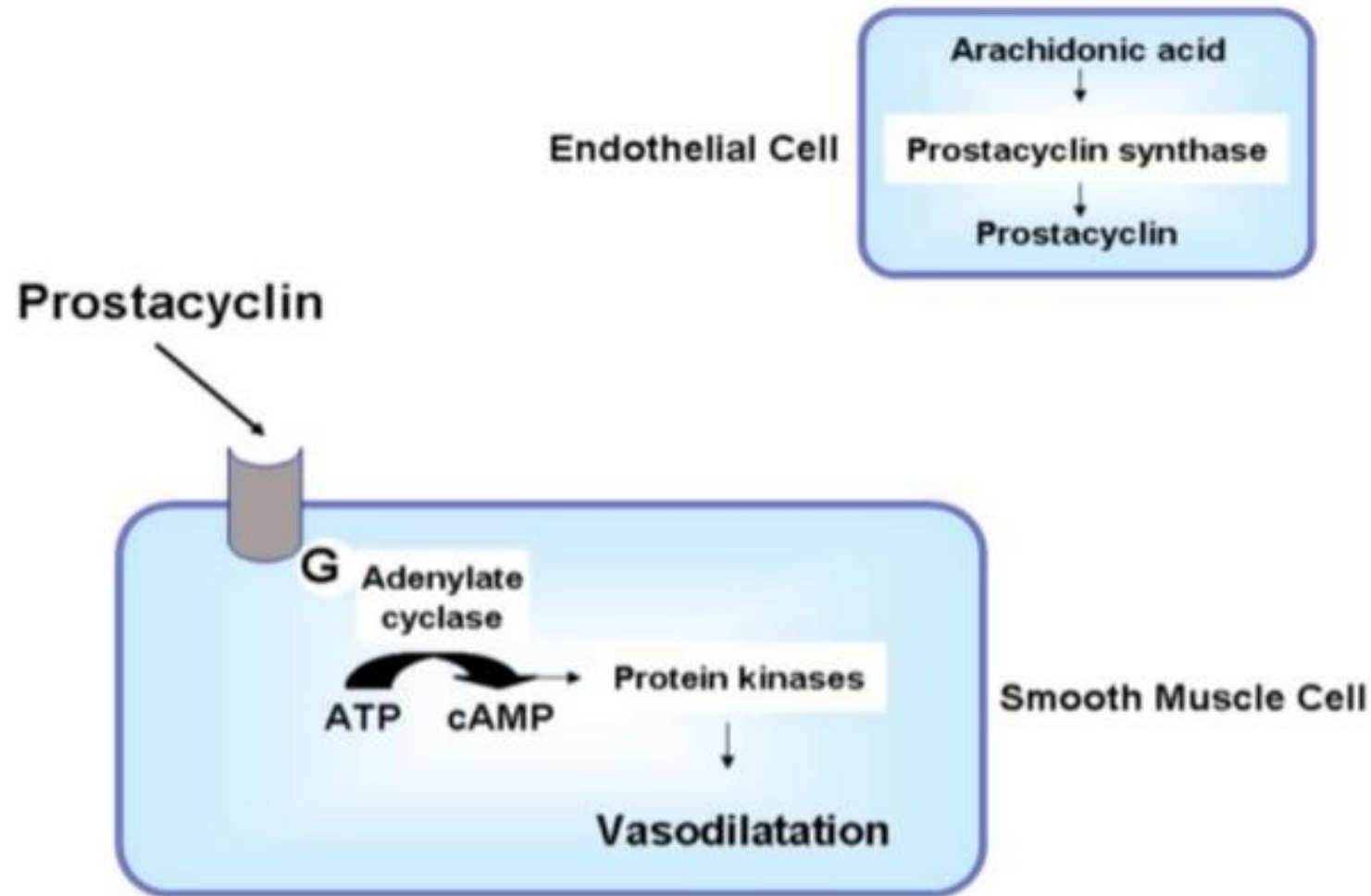


Prostacyclin causes VSMC relaxation

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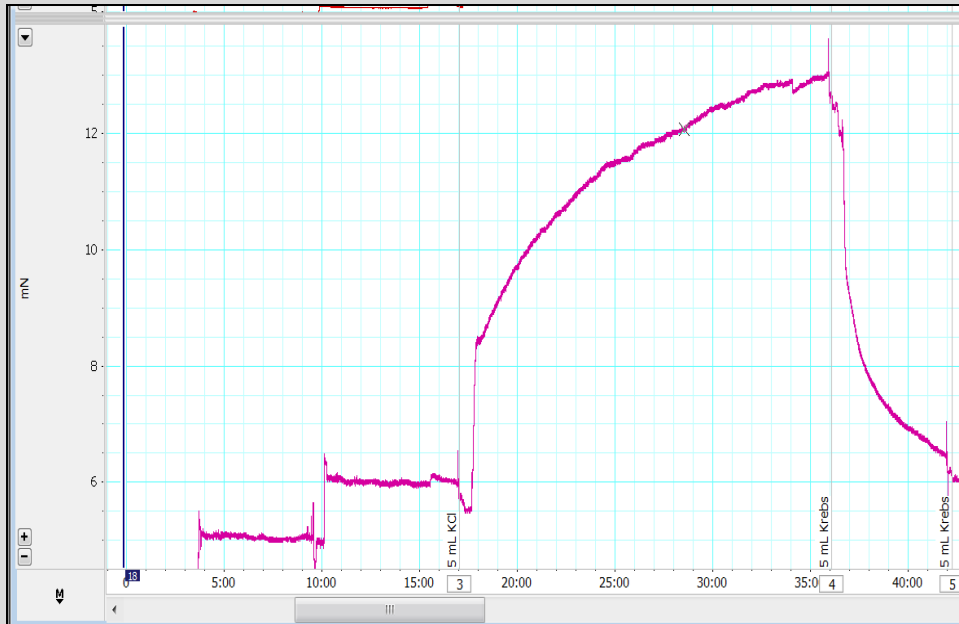
Prostacyclin Mechanism of Action



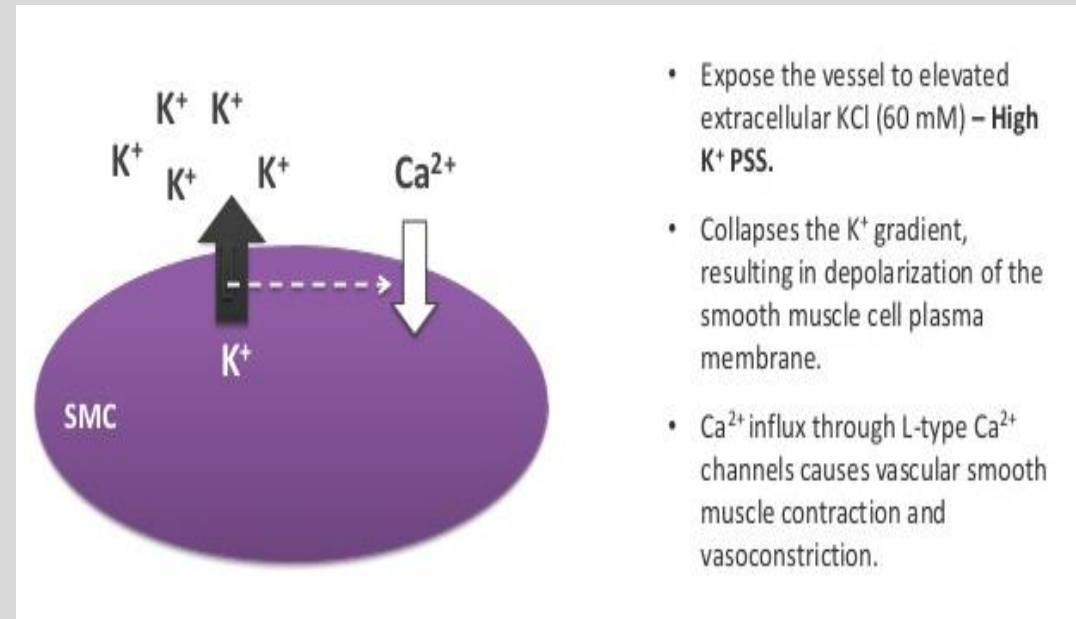
KCl causes VSMC contraction

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Sepulveda's lab – NYITCOM -Evan Asher.

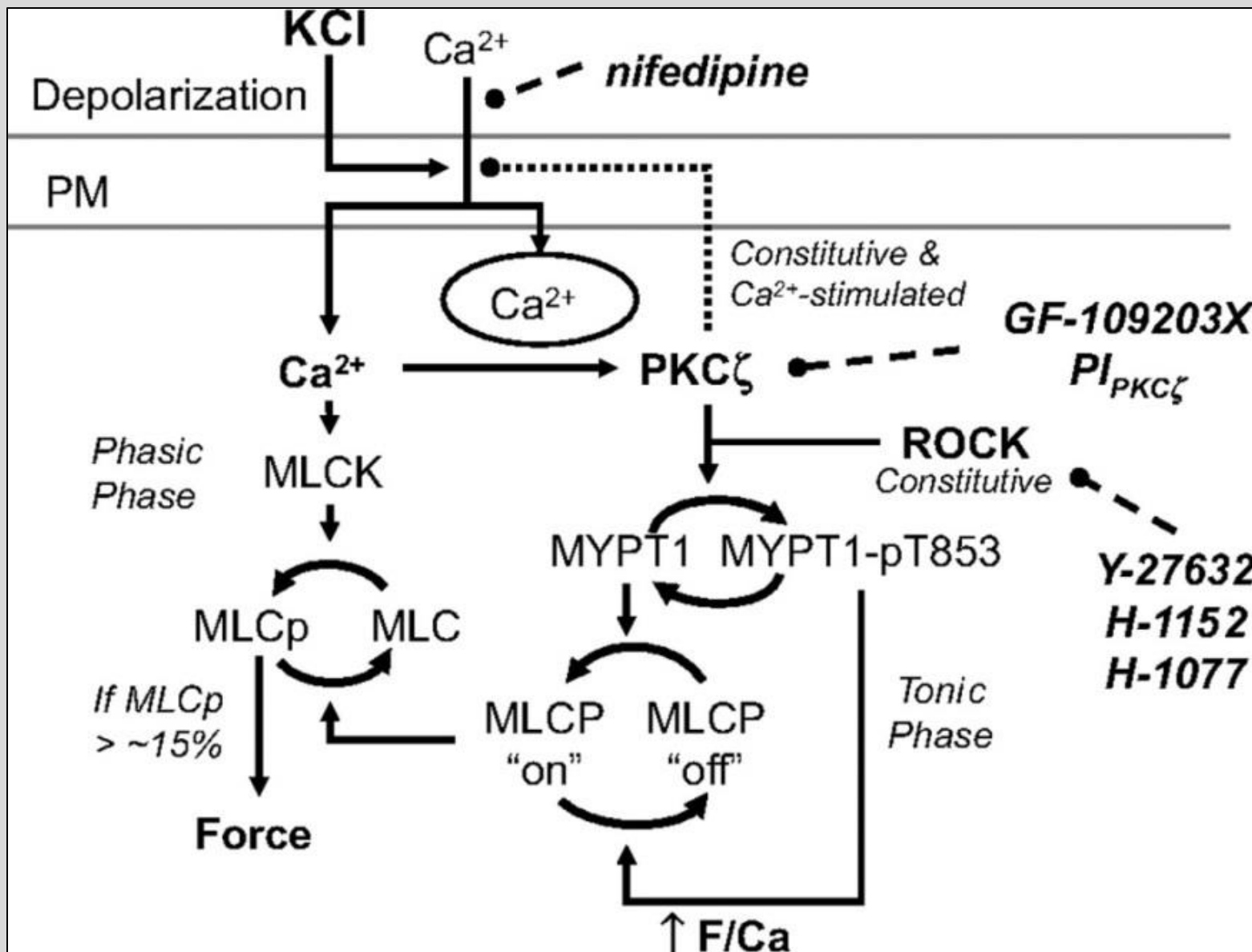


KCl → depolarization of the membrane potential → Ca^{++} signaling

KCl causes VSMC contraction

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- Nifedipine
 - Amlodipine
 - Verapamil
- All L-type calcium blocker

♥ Calcium plays a crucial role in contraction in **cardiac, smooth, and skeletal muscle**

♥ Myocyte depolarization causes activation of L-type calcium channels on the plasma membrane

♥ In cardiac and smooth muscle cells, myocyte undergoes depolarization which leads to activation of L-type calcium channels on the plasma membrane and influx of extracellular calcium.

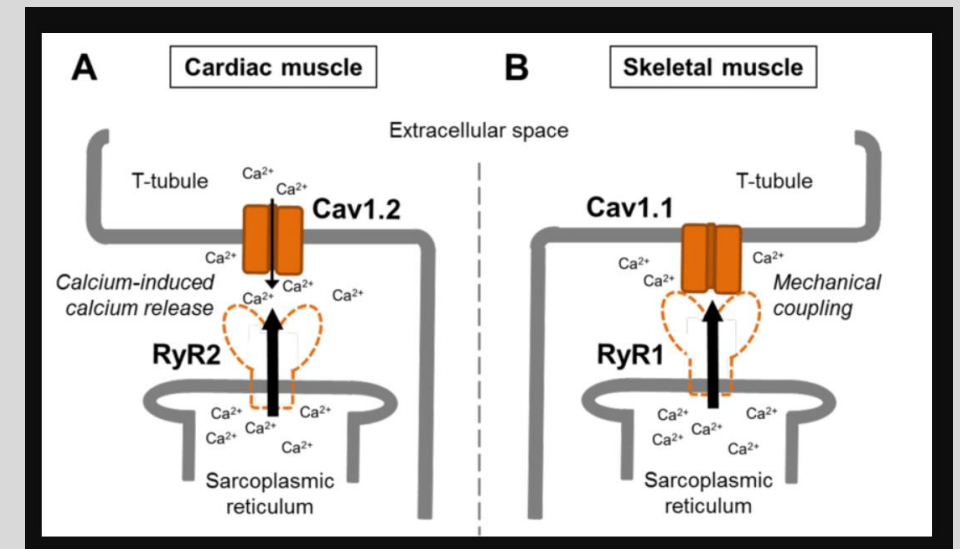
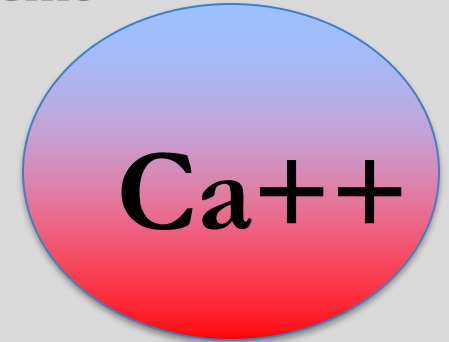
♥ Calcium binds to troponin (cardiac muscle) or calmodulin (smooth muscle), allowing actin and myosin to interact and cause muscle contraction

♥ In contrast, the L-type calcium channels in skeletal muscle directly interact with RyR calcium channels to release calcium from the sarcoplasmic reticulum. This mechanism occurs without significant influx of calcium across the plasma membrane

♥ Because skeletal muscle is not dependent on extracellular calcium influx, channels blockers do not affect skeletal muscle contractility

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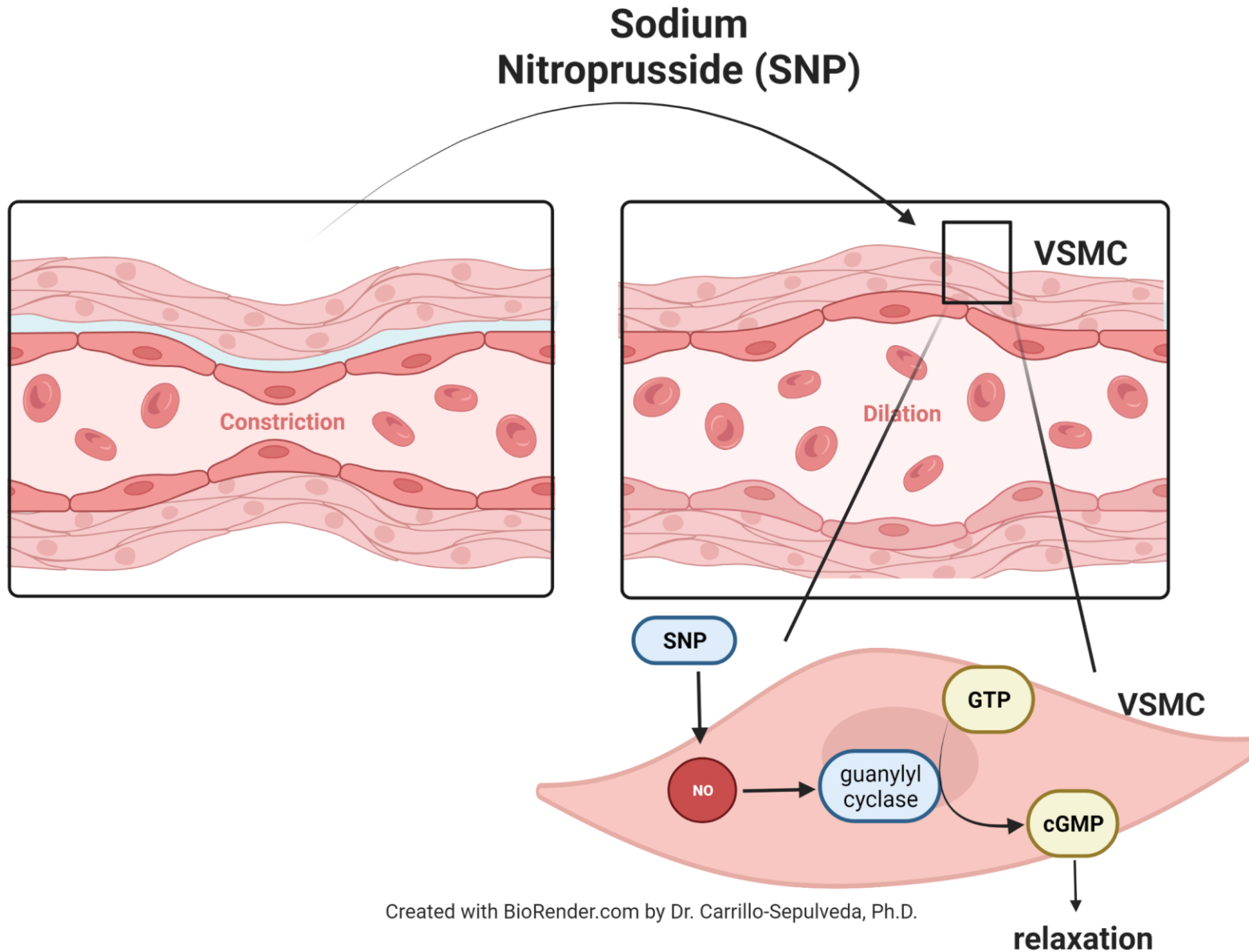
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VSMC Relaxation

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Summary

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- Contractile proteins involved in the VSMC contraction
- Mechanism of VSMC contraction: calcium plus calmodulin complex
- Effects of sodium nitroprusside and KCL in VSMC

Contacts

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- Questions: alicia.sepulveda@nyit.edu

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